OO-D-691F
August 17, 1995
SUPERSEDING
OO-D-691E
May 1, 1975

FEDERAL SPECIFICATION

DRILL, PNEUMATIC, PORTABLE: AND BORER, WOOD, PNEUMATIC

The General Services Administration has authorized the use of this Federal Specification by all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers portable pneumatic drills and pneumatic wood borers for drilling holes in mild steel and in wood.

1.1.1 Federal specification coverage. Federal specifications do not cover all varieties of the commodity indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Types, and styles. Portable pneumatic drills and pneumatic wood borers shall be of the following types and styles, as specified (see 6.2):

Type I - Drills (Reversible and Nonreversible)
Style A - 90° Angle Head
Style B - 360° Angle Head
Style C - Straight Drive

Type II - Wood Borers

Type III - Corner Drills

1.2.2 Sizes. Drills and borers covered by this specification shall be of the sizes shown in the respective tables, as specified (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

PPP-P-40 - Preservation and Packing of Hand Tools and Tool Accessories for Power Driven, Metal and Woodworking Machinery

Federal Standards:

FED-STD-H28 - Screw-Threads Standards for Federal Services
FED-STD-H28/7 - Screw-Thread Standards for Federal Service, Section 7 Pipe Threads - General Purpose

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2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American National Standard Institute (ANSI):

ANSI B5.10	-	Machine Tapers
ANSI B107.4	-	Driving and Spindle Ends for Portable Hand Air and Electric Tools
ANSI S5.1	-	Test Code for the Measurement of Sound from Pneumatic Equipment

(Application for copies should be addressed to the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.)

American National Standards Institute (ANSI)/American Society of Quality Control (ASQC):

ANSI/ASQC Z1.4	-	Sampling Procedures and Tables for Inspection by Attributes

(Application for copies should be addressed to the American Society for Quality Control, P.O. Box 3005, 611 E. Wisconsin Avenue, Milwaukee, WI 53201-4606.)

Compressed Air and Gas Institute

CAGI-PNEUROP	-	Test Code for the Measurement of Sound from Pneumatic Equipment

(Application for copies should be addressed to the Compressed Air and Gas Institute, 1300 Sumner Avenue, Cleveland, OH 44115.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

3. REQUIREMENTS

3.1 Illustrations. Illustrations herein are descriptive and not restrictive and are not intended to preclude the purchase of drills or borers otherwise conforming to this specification.

3.2 First article sample. When specified (see 6.2), a completely fabricated sample shall be subjected to first article testing in accordance with 4.3. The sample shall be a preproduction item or a production item that meets the requirements of this specification. The sample shall be identical with the proposed items and shall have been made by the methods that are to be used in a production of items to fulfill the terms of the contract. Approval of the sample under first article

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inspection shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements of this specification.

3.3 Material. The materials used in the construction of all components shall be capable of completing the examinations and tests specified herein. All components shall be new and unused. Repaired or reconditioned components shall not be acceptable.

3.4 Interchangeability. All parts having the same manufacturer's part number shall be constructed to definite standards, tolerances, and clearance in order that such parts may be completely interchangeable and may be replaced or adjusted without requiring modification.

3.5 Screw threads. All threaded parts, including screws, bolts, and nuts, shall conform to the applicable requirements of FED-STD-H28.

3.6 Identification markings. Each drill and borer shall be marked in a permanent and legible manner showing at least the following: the manufacturers' name or registered trade mark, model number, size, and maximum no load speed.

3.7 Lubrication.

3.7.1 Primary lubrication. The 1/4 and 3/8-inch size drills shall be lubricated in accordance with the manufacturer's instructions. The 1/2-inch and larger sizes of drills and borers shall have a built-in oil reservoir of a capacity which shall insure metered lubrication of all moving parts in the air stream from one filling for a minimum operating period of eight hours.

3.7.2 Secondary lubrication. Drills and borers requiring periodic lubrication of parts other than those in the air stream shall be provided with external means of applying the required lubricant.

3.8 Air inlet connection. The air inlet connection shall consist of a well secured brass or ferrous fitting threaded with internal American standard taper pipe threads (NPT) or internal dryseal American standard taper pipe threads (NPTF) in accordance with FED-STD-H28/7. The air inlet shall be fitted with an air strainer or screen (20 mesh or finer) which shall effectively retain solid particles in the compressed air supply. The strainer or screen shall be of such design that it can be easily removed for cleaning.

3.9 Bearings. Except as otherwise specified herein, all rotating parts shall be provided with commercially available bearings. Roller bearings, which are designed primarily for radial loads, shall not be used in a thrust position. Open type (needle) bearings shall be suitably housed and sealed to prevent leakage of lubricant and entrance of dirt and dust. If sleeve bearings are used, they shall be of the sintered, metal powder, or oil-impregnated type.

3.10 General construction. Drills and borers shall be of rugged construction so as to withstand, without failure, treatment likely to be encountered under general service conditions. Drills and borers shall be self contained, readily accessible for adjustment or replacement of parts. Parts which require lubrication shall be properly enclosed so as to prevent entrance of foreign particles and leakage of lubricant. Working parts exposed to wear shall be of sufficient hardness to withstand the service required. Parts shall be properly finished by machining or grinding to dimensions within limits established by good commercial practice. Drills and borers shall be reliable and effective when operated on compressed air lines at gage pressures between 80 and 100 pounds per square inch gage (psig). Reversible drills and borers shall be furnished with a conveniently located mechanism for reversing the rotation of the spindle. The rotational control shall be plainly and permanently marked to indicate direction of rotation.

3.11 Housing. Strength of housing shall be such to prevent breakage or indentation under service conditions. When threaded fasteners are utilized to assemble adjacent housing components, through bolting shall be used in preference to tapped holes. Cap screws, if used, shall engage threads in ferrous housings or ferrous inserts in nonferrous housings by not less than a length equal to the pitch diameter of the fastener thread. Where ferrous inserts are not used in nonferrous housings, cap screws shall engage threads by not less than a length equal to 1-1/2 times the pitch diameter of the fastener thread. All threaded fasteners shall be secured with suitable locking devices.

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3.12 Motor. The motor shall be of the rotary vane type. The rotor of the motor shall consist of one or more elements accurately balanced with the rotor shaft and shall be propelled by compressed air through the attendant action of vanes or blades made of suitable composition.

3.13 Speed. The speed of the drills and borers shall be consistent with the requirements stated herein. There shall be no abrupt or erratic speed variations affecting drilling and boring operations when tested in accordance with 4.7.1.

3.14 Handles. The handles of drills and borers shall be as specified herein.

3.15 Sound pressure levels. The sound pressure levels measured at an unobstructed distance of one meter, shall not exceed the following limits on the A weighted scale in any prescribed microphone location (see 4.7.2):

90 dBA - Type I, styles A, B, and C in sizes 1/2 inch and smaller
100 dBA - Type I, styles A, B, and C in sizes 7/8 inch and larger; type II; and type III

3.16 Throttle. The throttle for all drills and borers shall be of the type specified herein, control the speed of the tool, and automatically shut off the air supply when released. A lock-on device may be provided if turnoff can be accomplished by a single motion of the same finger or fingers that actuated it.

3.17 Spindle. The spindle of the drills and borers shall be of the type and size specified herein and shall conform to ANSI B5.10 or B107.4 as applicable.

3.18 Type I drills. Type I drills shall be suitable for drilling holes in mild steel with the size high speed twist drill of a diameter corresponding to the size of the drill. The drills shall be of the style and size as specified, (see 6.2), be nonreversible, and shall conform to the requirements in Tables I, II, and IV. When specified (see 6.2), drills shall be reversible and shall conform to the requirements of Tables I, III, and IV.

TABLE I. Type I drills, sizes, and capacities

Size (Inches)	At Spindle	Air Inlet Connection Nominal Pipe Size (Inches)	Weight-pounds (Max.)
1/4 High Speed	1/4-inch chuck 1/	1/4	4-1/4
1/4 Medium Speed	1/4-inch chuck 1/	1/4	6
1/4 Low Speed	1/4-inch chuck 1/	1/4	6
3/8 High Speed	3/8-inch chuck 1/	1/4	8
3/8 Medium Speed	3/8-inch chuck 1/	1/4	8
3/8 Low Speed	3/8-inch chuck 1/	1/4 or 3/8	8
1/2 Pistol Grip	1/2-inch chuck 1/	1/4 or 3/8	8-1/2
1/2 Heavy Duty	No. 2 ASA Morse Taper Socket	1/2 or 3/8	18 2/
7/8	No. 3 ASA Morse Taper Socket	1/2 or 3/4	28
1-1/4	No. 4 ASA Morse Taper Socket	1/2 or 3/4	40
2	No. 4 ASA Morse Taper Socket	3/4 or 1	45
3	No. 5 ASA Morse Taper Socket	3/4 or 1	70

1/ For type of spindle on which chuck is mounted, see 3.18.4.
2/ If furnished with chuck, add 2 pounds.

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TABLE II. Type I, style C (nonreversible) drills - performance characteristics

Size (Inches)	Specified Torque (Ft-lbs)	At Specified Torque		Speed At No Load (Max.) (RPM)
		Speed (Min.) (RPM)	Air Consumption (Max.) (scfm/hp)	
1/4 High Speed	0.75	1400	75	3200
1/4 Medium Speed	2.0	950	55	2300
1/4 Low Speed	2.6	500	55	1000
3/8 High Speed	2.5	650	70	1600
3/8 Medium Speed	6.0	450	60	900
3/8 Low Speed	15	175	60	350
1/2 Pistol Grip	8	400	45	1200
1/2 Heavy Duty	10	500	40	1000
7/8	30	300	30	500
1-1/4	50	260	35	350
2	80	150	35	310
3	200	100	30	150

TABLE III. Type I, style C (reversible) drills - performance characteristics

Size (Inches)	Specified Torque (Ft-lbs)	At Specified Torque		Speed At No Load (Max.) (RPM)
		Speed (Min.) (RPM)	Air Consumption (Max.) (scfm/hp)	
1/4 High Speed	0.75	1250	85	3000
1/4 Medium Speed	2.0	800	70	1900
1/4 Low Speed	2.6	300	55	900
3/8 High Speed	2.5	550	70	1200
3/8 Medium Speed	6.0	350	40	800
1/2 Pistol Grip	8	400	35	1200
1/2 Heavy Duty	10	500	40	800
7/8	30	375	45	500
1-1/4	50	225	40	350
2	80	150	50	310
3	200	75	40	125

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TABLE IV. Type I, styles A and B drills - performance characteristics

Size (Inches)	Specified Torque (Ft-lbs)	At Specified Torque		Speed At No Load (Max.) (RPM)
		Speed (Min.) (RPM)	Air Consumption (Max.) (scfm/hp)	
Style A, 90° Angle Head (Nonreversible)				
1/4 High Speed	0.75	1400	70	3500
1/4 Medium Speed	2.0	700	60	2400
3/8 High Speed	2.5	700	60	2000
3/8 Medium Speed	6.0	450	50	1600
Style A, 90° Angle Head (Reversible)				
1/4 High Speed	0.75	1250	80	3000
1/4 Medium Speed	2.0	600	70	2200
3/8 High Speed	2.5	700	60	1600
3/8 Medium Speed	6.0	350	50	1200
Style B, 360° Angle Head (Nonreversible)				
1/4 High Speed	1.5	1200	55	3400

3.18.1 Style A, 90° angle head. Style A drills shall have a 90° angle head, body grip handle, lever type throttle, be similar to Figure 1, and shall conform to the requirements of Tables I and IV, as applicable.

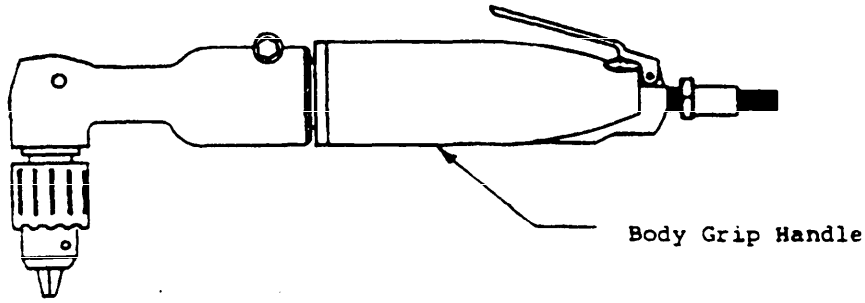


Figure 1. Type I, style A, 90° angle head drill.

3.18.2 Style B, 360° angle head. Style B drills shall have an adjustable 360° angle head, body grip handle, lever type throttle, be similar to Figure 2, and shall conform to the requirements of Tables I and IV.

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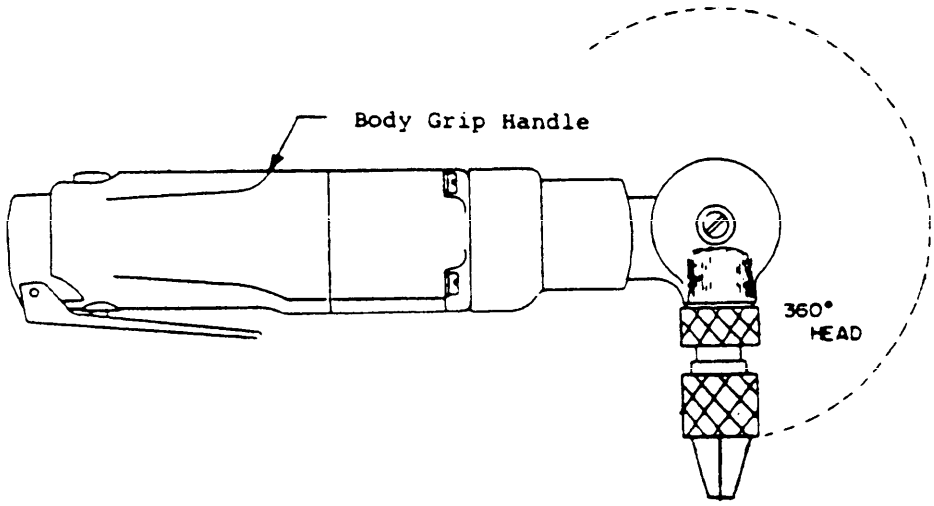


Figure 2. Type I, style B, 360° angle head drill.

3.18.3 Style C, straight drive. Style C drills shall be similar to Figures 3 or 4 and shall conform the requirements of Tables I, II, and III, as applicable.

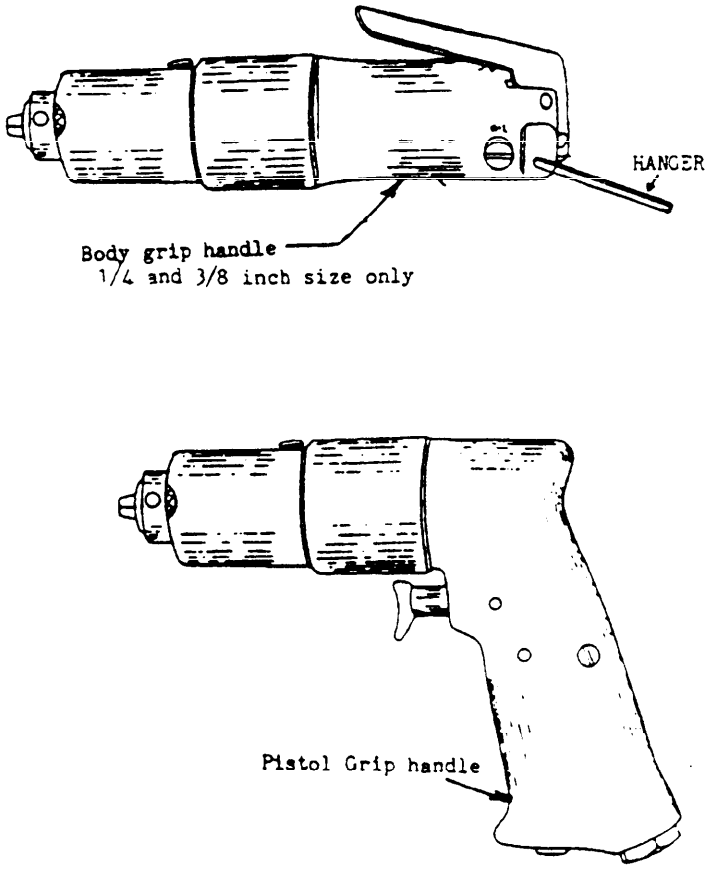


Figure 3. Type I, style C, straight drive drill, 1/4, 3/8, and 1/2-inch sizes.

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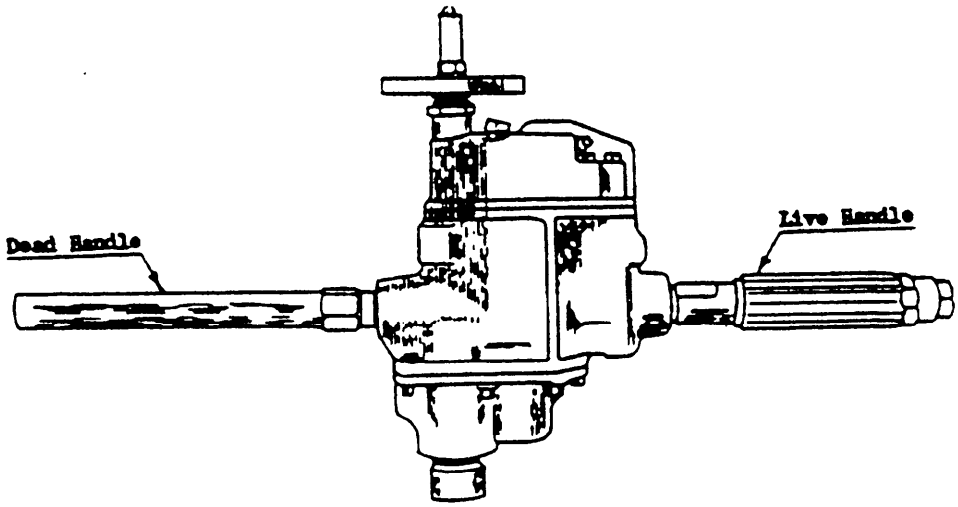


Figure 4. Type I, style C, straight drive drill, 1/2 inch (heavy duty) and larger sizes.

3.18.3.1 Style C drill, 1/2-inch (heavy duty). Style C drill, 1/2-inch (heavy duty) size and larger sizes shall be similar to Figure 4, shall have a "live" handle equipped with a lever or sleeve style throttle, a removable "dead" handle and a feed screw. The handles shall be secured to opposite sides of the housing and be perpendicular to the spindle axis. The live handle shall have an air inlet connection of the size specified in Table I. When specified (see 6.2), the 1/2-inch (heavy duty) size shall have a spade handle in lieu of a feed screw and it shall be located at the end of the housing opposite the chuck and suitably in line with the longitudinal axis of the chuck.

3.18.3.1.1 Feed screw. The style C drill, 1/2-inch (heavy duty) size and larger sizes shall have a feed screw having four radiating arms for hand feed control. When specified (see 6.2), these drills may be furnished without a feed screw. The feed screw shall be of such design that it cannot be unintentionally screwed entirely out of the body of the drill. The feed screw shall be of such construction that there will be no unsteadiness when the drill is operated with the feed screw at its maximum extension. The end of the feed screw projecting into the body of the drill shall be fitted with an ejector for forcing tools out of the socket by turning the feed screw handle. Drills equipped with a socket external to the body of the drill shall have tool ejection accomplished by means of a driftpin. The outer or external end of the feed screw shall be fitted with a hardened steel center.

3.18.3.2 Style C drill, 1/2-inch (pistol grip). The style C drill, 1/2-inch (pistol grip) size and smaller sizes shall be similar to Figure 3 and shall have a pistol grip handle, and a trigger or button type throttle. The 1/2-inch (pistol grip) size shall have a removable dead handle having at least two positions, one on each side of the housing, perpendicular to the spindle axis. When specified (see 6.2), the 1/4 and 3/8 inch size shall have a body grip handle with a lever style throttle.

3.18.4 Spindles for style A and C drills.

3.18.4.1 Sizes 1/4, 3/8, and 1/2-inch (pistol grip). Spindles for the 1/4, 3/8, and 1/2-inch (pistol grip) size drills in accordance with Figures 1 and 3 shall conform to the data specified in Table V.

3.18.4.2 Sizes 1/2-inch (heavy duty) and larger. Spindles for the 1/2-inch (heavy duty) and larger size drills shall have an ASA Morse taper Socket of the size specified in Table I.

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TABLE V. Spindle sizes for type I, styles A and C drills

Sizes (Inches)	Nonreversible	Reversible
1/4	3/8-24 Thread	No. 1 Male Jacob Taper, or 3/8-24 Right Hand Thread with #10-32 Left Hand Lock Screw
3/8	3/8-24 or 1/2-20 Thread	No. 2 Male Jacob Taper, or 3/8-24 or 1/2-20 Right Hand Thread with #10-32 Left Hand Lock Screw
1/2 (Pistol Grip)	1/2-20, 5/8-16, or 45/64-16 Thread	No. 3 Male Jacob Taper

3.18.5 Chucks.

3.18.5.1 Styles A and C drills. The style A and C drills in accordance with Figures 1 and 3 shall be equipped with a three jaw, geared type, key operated medium duty (commercially rated) chuck. The chuck shall accommodate a straight twist drill of the size corresponding to the size of the drill and shall fit directly on the spindle. When specified (see 6.2), the 1/2-inch (heavy duty) size in accordance with Figure 4 shall be equipped with the same kind of chuck. A chuck key shall be furnished for each chuck.

3.18.5.2 Style B drills. The style B drills shall be equipped with a keyless, jaw type clutch that is capable of accommodating straight shank twist drills up to 1/4 inch in diameter. Suitable means shall be provided for manual operation of the chuck.

3.19 Type II wood borers. Wood borers shall be suitable for boring holes in dry, well seasoned oak hardwood with the maximum size of wood bit, in inches, corresponding to the rated size of the wood borer, when operated at an air pressure of 90 psig. Wood borers shall be of the size specified (see 6.2), be similar to Figure 5 and conform to the requirements specified in Table VI. The borers shall be reversible and conform to the performance requirements when operating in the forward rotational direction.

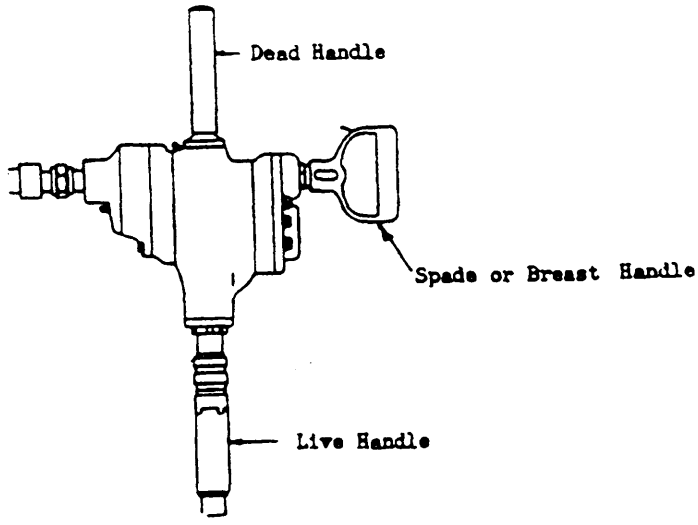


Figure 5. Type II, wood borers.

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TABLE VI. Type II, wood borers: Dimensions and performance characteristics

Size (Inches)	Chuck Size for Woodbit Shanks (Dia.) (Inch)	Horsepower (Min.)	Air Consumption (Max.) scfm/ HP-40	Air Inlet Connection Nominal Pipe Size (Inch)	ASA Morse Taper Socket in Spindle (Number)	Weight (Max.) (Pounds)	Speed At No Load (Max.) (RPM)
1	1/2	1.0	40	1/2	2	17	1000
2	1/2	1.5	40	1/2 or 3/4	2	35	850
4	1/2	2.0	40	1/2 or 3/4	3	35	400

3.19.1 Handles. Wood borers shall be equipped with a "live" handle, a "dead" handle, and a spade or breast handle. The "live" and "dead" handles shall be secured to opposite sides of the housing, perpendicular to the spindle axis. The "live" handle shall be provided at its outer end with an air inlet connection on nominal pipe size, as specified in Table VI. The "dead" handle shall be removable. The spade or breast handle shall be located at the end of the housing opposite the chuck and suitably in the line with the axis of the chuck.

3.19.2 Throttle. Wood borers shall be provided with lever or sleeve style throttle on the "live" handle.

3.19.3 Spindle. Wood borers shall be equipped with spindles having standard ASA (Morse) taper sockets of the sizes shown in Table VI, threaded externally on its outer end in accordance with the manufacturer's standard practice, and shall be furnished with chucks suitable for retaining shanks of the sizes shown in Table VI.

3.20 Type III corner drills. Corner drills shall be suitable for use in close places where an ordinary drill cannot be operated. Corner drills shall be constructed with the spindle at one end and the motor at the opposite end of the housing. Corner drills shall be similar to Figure 6 and shall conform to the dimensions in Table VII for the size specified (see 6.2). Corner drills shall be reversible and shall conform to the performance characteristics specified in Table VIII, in both directions and shall be furnished with a feed screw.

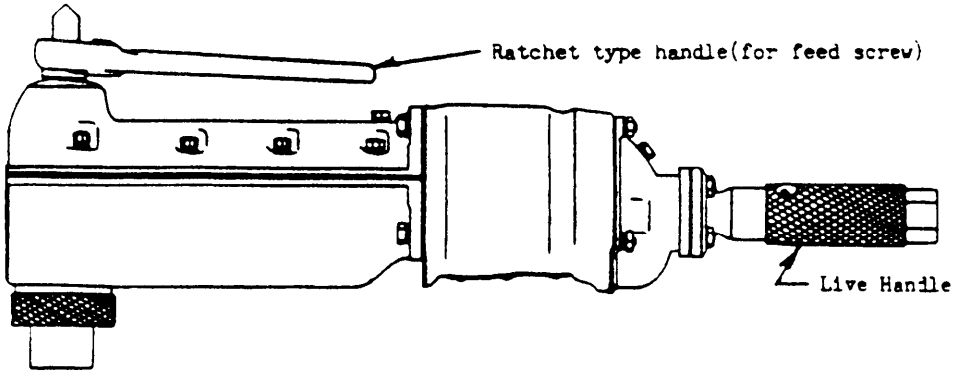


Figure 6. Type III, corner drills.

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TABLE VII. Type III corner drills - dimensions

Size	ASA Morse Taper Socket in Spindle (Number)	Air Inlet Connection Nominal Pipe Size (Inch)	Weight (Max.) (Pounds)	Approach (Max.) (Inches)
7/8	3	1/2 or 3/4	40	1-9/16
1-1/4	4	1/2 or 3/4	50	1-9/16
3	5	1/2 or 3/4	60	2-3/8

TABLE VIII. Type III corner drills - performance characteristics

Size (Inches)	Specified Torque (Ft-lbs)	At Specified Torque		Speed at No Load (Max.)
		Speed (Min.) (RPM)	Air Consumption (Max.) (scfm/hp)	
7/8	100	180	35	300
1-1/4	160	100	30	300
3	250	70	30	150

3.20.1 Handle. Corner drills shall be equipped with a "live" handle suitably in line with the longitudinal axis of the air motor and at the opposite end of the housing from the spindle. The handle shall be provided at its outer end with an air inlet connection of nominal pipe size, as specified in Table VII.

3.20.2 Throttle. Corner drills shall be equipped with a lever or sleeve style throttle on the "live" handle.

3.20.3 Spindle. Corner drills shall be equipped with spindles having standard ASA Morse taper sockets of the sizes shown in Table VI.

3.20.4 Feed screw. Feed screws furnished with the corner drills shall conform to 3.18.3.1.1 except that the feed screw shall be actuated by a reversible ratchet-type handle or other suitable mechanism, operable within the width of the drill body, for hand-feed control.

3.21 Accessories or extra equipment. When accessories or extra equipment are required, they shall be as specified (see 6.2).

3.22 Technical manual. One copy of a technical manual consisting of the manufacturer's standard commercial instruction and parts list, shall be furnished with each drill and borer.

3.23 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the specification preparing activity for change to this document.

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3.24 Workmanship. Details of workmanship shall be in accordance with the best commercial standards and practices. Paints, coatings, platings, and finishes shall be smooth, dry, adherent, continuous, and not stained or discolored. Fasteners shall be firmly secured and show no evidence of deformation, cross threading, or hazardous burrs. Adhesives and lubricants adequate for their intended purpose shall be properly and neatly applied. Adhesives shall be adequately cured. There shall be no interference, binding, or galling of parts. External and bearing surfaces shall be free of tool and gouge marks, nicks, or other surface imperfections. The item shall be clean and free of corrosion and debris (e.g., chips, shavings, slivers) or other foreign material. The item shall be free from manufacturing workmanship defects (e.g., loose, missing, binding or misaligned parts, sharp or rough external edges, corners or surfaces) and material workmanship defects (e.g., pits, rips, fins, burrs, tears, nodules, cracks, blisters) which may adversely impact the item's serviceability, durability, safety, or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. In accordance with 4.1 above, the supplier is responsible for insuring that components and materials used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specification and standards to the extent specified, or, if none, in accordance with this specification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be applied to the sample submitted in accordance with 3.2. First article inspection shall consist of the examinations specified in 4.6, the tests specified in 4.7, and the test sequence specified in 4.7.1. Failure of the sample to pass the examinations and tests shall be cause for rejection.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of all the examinations specified in 4.6 and the performance tests specified in 4.7.1.

4.4.1 Inspection lot. An inspection lot shall consist of all the drills or borers of the same type, style, and size offered at the same time any place for quality conformance inspection.

4.5 Sampling procedures. Unless otherwise specified, sampling for inspection shall be performed in accordance with ANSI/ASQC Z1.4.

4.6 Examination.

4.6.1 Visual examination. The end items shall be examined for the defects listed in Table IX. The lot size shall be expressed in terms of drills. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of percent defective, shall be 1.5 for total (major and minor combined) defects.

4.6.2 Dimensional examination. The end items shall be examined for any nonconformance to dimensional requirements. The lot size shall be expressed in terms of drills. The sample unit shall be one drill. The inspection level shall be II and the AQL, expressed in percent defective shall be 1.5 for total (major and minor combined) defects.

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TABLE IX. Classification of defects

Examine	Defects Critical	Major	Minor
Type, style, or size, not as specified			101
Style of handle, not as specified			102
Drill or borer incomplete, parts missing			103
Drill or borer, not securely assembled; fasteners, not secured with suitable locking device			104
Spindle, not as specified			105
Chuck, not as specified			106
Marking missing, illegible, or incorrect			107
Drill or borer, not provided with a means of applying lubrication			108
Air-inlet connection damaged or not as specified			109
Air strainer damaged, missing, or not easily removable			110
Handles not removable when required			111
Throttle not as specified			112
Material not as specified			113
Like parts for same manufacturer not interchangeable			114
Bearings not as specified or not adequately-protected from dirt and foreign matter			115
Workmanship not as specified			116
Tapped holes under minimum depth specified or not reinforced as applicable			117

4.6.3 Testing. The end items shall be tested for the performance requirements found in the applicable table(s). The lot size shall be expressed in units of drills. The sample unit shall be one drill. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 1.5.

4.7 Tests. All tests shall be conducted with an air supply of 90 psig (620 kPa) measured as close to the tool as practicable. The test tool shall be connected to the air supply line by means of a 10 foot length of air hose having the following hose size (inner diameter): 5/8 inch hose size for drill sizes 1/2 inch and smaller; 3/4 inch hose size for drill sizes 7/8 inch and larger.

4.7.1 Performance tests. The tools shall be tested for air consumption, speed, torque, and horsepower, as specified in the applicable Table. The tests shall also determine compliance with the throttle, rotational, and reversibility action, as applicable. For type II, (size 2 wood borers), after determining that the no load speed is within requirements, determine the maximum torque (T) in foot-pounds (ft-lbs) within the 400 to 700 revolutions per minute (rpm) range by loading the tool on a dynometer. Record the maximum torque and corresponding rpm and calculate the horsepower (hp) using the formula:

$$hp = \frac{rpm \times T \text{ (foot-pounds)}}{5252}$$

The calculated HP shall be 1.5 or greater and air consumption shall not exceed 40 SCFM/HP.

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4.7.2 Sound pressure level test. The sound pressure level test shall be conducted in accordance with the applicable provisions of the Compressed Air and Gas Institute's CAGI-PNEUROP Test Code for the Measurement of Sound from Pneumatic Equipment, except as modified herein. The acceptable limits shall be as specified in 3.15.

4.7.2.1 Acoustical calibration. The entire instrumentation system shall be calibrat before each test series allowing a minimum instrument warm-up period of 5 minutes if transistorized and 1/2 hour if vacuum tube. Recalibration shall be performed if any instrument adjustment is necessary during progress of the test.

4.7.2.2 Background sound level. The ambient noise level shall be at least 10 decibels (dB) less than the measurement taken with the tool running in all tool and microphone orientations. In addition, the sound produced by the energy absorbing device, used in the "on load" test, must be at least 10 dBA below the machine's own sound output.

4.7.3 Air leakage test. Drills and borers shall be tested for tightness. Drills and borers shall be connected to a system consisting of a 0.25 cubic foot air receiver, valving as necessary and a 0 - 300 psi range pressure gage. System shall be pressurized with air or nitrogen to 150 psi with throttle in the closed position and isolated for a period of 15 minutes. At the end of 15 minutes, maximum pressure drop permitted shall be no more than 50 psi.

4.7.4 Service tests. Drills or borers shall complete 150 hours of actual work under practical service conditions to determine adequacy and maintainability of the tool. When test under practical service conditions is not feasible, a simulated test may be performed of equal duration by using a life testing machine. Tool shall be loaded to specified torque on a prony brake or other load applying device. Tool shall be cycled on for 20 seconds and off for 10 seconds.

4.7.4.1 Lubrication test. Lubrication system shall be checked to insure conformance with 3.7. This test may be performed concurrently with the service test. Lubricant reservoir shall be inspected periodically to determine adequacy of lubrication system.

4.8 Test sequence. All drills and borers shall be subjected to the test of 4.6.1 through 4.6.3, followed by the service tests (see 4.6.4), and then again subjected to the test of 4.6.1. The tools shall then be disassembled and examined. Any evidence of broken, deformed, or excessively worn components shall be cause for failure.

4.9 Examination of preparation for delivery. An examination shall be made to determine compliance with the preservation, packaging, packing, and marking in accordance with PPP-P-40.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be in accordance with PPP-P-40. The level of preservation, packaging, and packing shall be A, B, or Commercial, as specified (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. Pneumatic drills covered by this specification are intended for drilling holes in metal, using twist drills. Wood borers are for use in drilling holes in wood, using wood bits. Maximum sound pressure levels have been set to insure against occupational noise-induced hearing losses. The tools covered by this specification are considered intermittent duty tools, used in short bursts of 2 minutes duration or less. The exposure time for tool sizes up to and including 1/2 inch is four hours. The larger sizes are considered to be less intermittent and have an exposure time of two hours.

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6.1 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type, style, and size, if applicable, of item required (see 1.2).
c. If reversible type I drill is required (see 3.18).
d. If spade handle is required on type I, style C, 1/2-inch (heavy duty drill (see 3.18.3.1).
e. If feed screw is required on type I, style C, 1/2-inch (heavy duty drill (see 3.18.3.1.1.).
f. If body grip handle with lever type throttle is required on type I, style C, 1/4 and 3/8 inch sizes (see 3.18.3.2.).
g. If three jaw, geared type, key operated medium duty (commercially rated) (chuck is required on type I, style C, straight drive, 1/2-inch (heavy duty drill (see 3.18.5.1).
h. Accessories or extra equipment, when required (see 3.18.5.1).
i. Applicable levels of preservation, packaging, and packing required (see 5.1).
j. First article required (see 4.3)
k. Level of preservation, packaging, and packing required (see 5.1).
l. Marking, if other than specified.

6.2 This specification reflects the requirements of OSHA, in effect as of the date of this specification.

MILITARY INTERESTS:

Army - GL
Navy - SH
Air Force - 99

Preparing Activity:

GSA - FSS

Review Activities:

Army - AV, ME

User Activities:

Army - CE
Navy - MC, YD

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